

Sandwich Bounds for Gramian and Hankel Spectra

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Abstract—We derive upper and lower bounds for the eigenvalues of the controllability and observability Gramians, as well as for the Hankel singular values, of continuous-time single-input single-output LTI systems with diagonalizable state matrix. The analysis employs a Cauchy matrix formulation that yields explicit, easily computable bounds in terms of the state matrix eigenvalues. The results provide insight into how the Gramian and Hankel spectra decay with the system size. Illustrative examples on structured systems show how the derived bounds are able to capture distinct spectral-scaling regimes.

I. INTRODUCTION

The controllability and observability Gramians are cornerstone tools for analyzing linear time-invariant (LTI) systems. Their eigenvalues describe how control and output energy are distributed across the state space: directions associated with large eigenvalues are easily excitable or observable, while those associated with small eigenvalues are weakly controllable or observable. Consequently, the Gramian eigenvalues provide compact, physically interpretable summaries of the “energetic” structure of LTI systems [13], [16], [21]. These spectral features underpin a broad range of applications, including control-energy analysis [18], [23], [3], state-estimation performance [22], information-transmission efficiency [4], and the learnability of LTI models [20].

Despite their central role, analytical understanding of the dependence of Gramian eigenvalues on system dimension and structural features is still limited. Most existing results address upper bounds for individual eigenvalues, particularly the smallest eigenvalue of the controllability Gramian [18], [17], [3], [25], while only a few works provide more refined characterizations through lower bounds or asymptotic expressions [24], [14], [5]. Broader analyses covering the full Gramian spectrum have also been developed [19], [2], [6], [7], but these results remain mostly confined to upper-bound descriptions, with (informative) lower bounds still lacking.

Beyond their role in energy-based controllability and observability analysis, characterizing the full spectrum of the Gramians is also fundamental for model reduction. The eigenvalues of the product of the controllability and observability Gramians define the (squared) Hankel singular values, which quantify how strongly each state direction contributes to the input-output behavior of the system. The Hankel singular values govern the approximation error of optimal model reduction techniques, thereby setting fundamental limits on the achievable accuracy of reduced-order models [11], [1].

This paper develops upper and lower bounds for the eigenvalues of the controllability and observability Gramians, as well as for the Hankel singular values, of continuous-time single-input single-output LTI systems with diagonalizable state matrix. The analysis is based on a Cauchy-matrix representation of the Gramians, which leads to explicit and easily computable bounds expressed directly in terms of the state matrix spectrum. These expressions elucidate how the Gramian and Hankel spectra scale with the system dimension, and their behavior is illustrated on structured examples that reveal distinct scaling regimes, ranging from exponential to polynomial decay as the system size increases.

While the derived upper bounds share some similarities with those reported in [2] and related studies, they are established via different analytical techniques. More importantly, the proposed lower bounds constitute, to the best of the author’s knowledge, the first explicit results of this nature, offering new insight and fundamental limits on the scaling of Gramian eigenvalues and Hankel singular values.

Notation. For a matrix $A \in \mathbb{C}^{n \times m}$, $A^\top$ denotes its transpose, A^* its conjugate transpose, and A_{ij} its (i, j) entry. If $A \in \mathbb{R}^{n \times n}$ has real eigenvalues, we denote its i -th eigenvalue by $\lambda_i(A)$ and, unless otherwise specified, assume they are ordered nonincreasingly, i.e., $\lambda_1(A) \geq \lambda_2(A) \geq \dots \geq \lambda_n(A)$. We also write $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$ for the largest and smallest eigenvalues of A , respectively. Given index sets P, Q , the submatrix of A with rows in P and columns in Q is written $A[P, Q]$; when $P = Q$, we simply write $A[P]$ for the corresponding principal submatrix. The trace of A is denoted $\text{tr}(A)$ and the spectral norm of A by $\|A\|$. We write $A \sim B$ to indicate that two square matrices A, B are similar. For a (row or column) n -dimensional vector a , $\text{diag}(a)$ denotes the diagonal matrix with diagonal entries given by a . $\text{tridiag}(a, b, c) \in \mathbb{R}^{n \times n}$ denotes the tridiagonal matrix with sub-, main-, super-diagonal entries a, b, c , respectively. For symmetric matrices $A, B \in \mathbb{R}^{n \times n}$, $A \succeq B$ denotes that $A - B$ is positive semidefinite. The symbol $e_i \in \mathbb{R}^n$ denotes the i -th canonical vector of $\mathbb{R}^n$. Finally, for a set S , $|S|$ denotes the cardinality of S and $\text{Re}(x)$ the real part of $x \in \mathbb{C}$.

II. PRELIMINARIES

In this work, we consider continuous-time, LTI single-input single-output (SISO) systems of the form:

$$\begin{cases} \dot{x}(t) = Ax(t) + bu(t), \\ y(t) = cx(t), \end{cases} \quad (1)$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}$, and $y(t) \in \mathbb{R}$ denote the state, input, and output of the system, respectively. The matrices

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$A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^{n \times 1}$, and $c \in \mathbb{R}^{1 \times n}$ are the system, input, and output matrices, respectively. The input-output behavior of the system is compactly described by its transfer function $G(s) = c(sI - A)^{-1}b$.

Throughout, we make the following assumption.¹

Assumption 1: The state matrix A is Hurwitz stable, i.e., all its eigenvalues have strictly negative real part, and diagonalizable. In addition, the pair (A, b) is controllable and the pair (A, c) is observable.

The (infinite-horizon) *controllability Gramian* associated with system (1) is defined as

$$\mathcal{W} = \int_0^\infty e^{At} b b^\top e^{A^\top t} dt, \quad (2)$$

and it is positive definite if and only if the system is controllable. The controllability Gramian $\mathcal{W}$, and in particular its inverse, quantifies the input energy required to drive the system state from the origin to a prescribed configuration [3].

Dually, the (infinite-horizon) *observability Gramian* is

$$\mathcal{O} = \int_0^\infty e^{A^\top t} c^\top c e^{At} dt, \quad (3)$$

which is positive definite if and only if the system is observable. The observability Gramian quantifies how the initial state influences the output energy of the system in the absence of external inputs, see e.g. [8].

Scalar quantities derived from the eigenvalues of $\mathcal{W}$ and $\mathcal{O}$ provide compact measures of the dynamic properties of the system: the spectrum of $\mathcal{W}$ reflects how much input energy is required to move the state in different directions, whereas the spectrum of $\mathcal{O}$ indicates how strongly those directions influence the output signal.

The *Hankel operator* of an LTI system maps past inputs to future outputs. For the system (1), it is given by

$$(\mathcal{H}u_-)(t) = c e^{At} \int_{-\infty}^0 e^{-A\tau} b u_-(\tau) d\tau, \quad t > 0,$$

where u_- denotes the past input ($t \leq 0$). For systems that are controllable, observable, and stable, $\mathcal{H}$ is compact and its nonzero singular values, the *Hankel singular values*, quantify the joint controllability-observability energy of each state direction. Such singular values satisfy [1]

$$\sigma_i = \sqrt{\lambda_i(\mathcal{W}\mathcal{O})}, \quad i = 1, \dots, n, \quad (4)$$

where $\mathcal{W}$ and $\mathcal{O}$ are the controllability and observability Gramians defined in (2)-(3).

The Hankel singular values play a central role in model reduction, particularly in *balanced truncation* [1]. In balanced coordinates, the controllability and observability Gramians coincide and are diagonal,

$$\mathcal{W} = \mathcal{O} = \text{diag}(\sigma_1, \dots, \sigma_n), \quad (5)$$

so that each state contributes independently with energy level σ_i . States with small σ_i can be truncated, yielding a

¹While Hurwitz stability, controllability, and observability are standard system-theoretic assumptions, the diagonalizability of A is here introduced to simplify the analysis of Gramians.

reduced model of order $r < n$, described by transfer function $\tilde{G}(s)$, that preserves the dominant input-output behavior. Approximation quality is measured by the standard system norms, e.g., the $\mathcal{H}_\infty$ norm:

$$\|G\|_{\mathcal{H}_\infty} := \sup_{\omega \in \mathbb{R}} \|G(i\omega)\|,$$

where $\|\cdot\|$ denotes the spectral norm. For the balanced truncation model $\tilde{G}$ of order r , the approximation error satisfies the classical bounds [1]

$$\sigma_{r+1} \leq \|G - \tilde{G}\|_{\mathcal{H}_\infty} \leq 2 \sum_{i=r+1}^n \sigma_i.$$

Thus, the tail of the Hankel singular values provides lower and upper control of the balanced truncation error.

The connection between Hankel singular values and approximation limits is even more direct in the case of *optimal Hankel-norm approximation*. The *Hankel norm* of a stable system is

$$\|G\|_{\text{Hankel}} := \sqrt{\lambda_{\max}(\mathcal{W}\mathcal{O})} = \sigma_1,$$

i.e., the largest Hankel singular value. A classic result states [11]

$$\min_{\tilde{G}, \text{order}(\tilde{G}) \leq r} \|G - \tilde{G}\|_{\text{Hankel}} = \sigma_{r+1}. \quad (6)$$

Hence, in this context, the Hankel singular values exactly characterize the attainable reduction error.

III. SANDWICH BOUNDS ON GRAMIAN EIGENVALUES

In the following, we focus on the controllability Gramian $\mathcal{W}$; however, all results apply to the observability Gramian $\mathcal{O}$ after suitable adaptations.

We begin with a result that establishes an explicit relationship between the controllability Gramian and a Cauchy matrix determined by the eigenvalues of A .

Lemma 1: (Gramian and Cauchy matrix) Consider the system (1). Let $\{\lambda_i\}_{i=1}^n$ denote the eigenvalues of A and V the eigenvector matrix of A . Define $Q := V \text{diag}(V^{-1}b)$. Then:

(i) The controllability Gramian $\mathcal{W}$ admits the factorization

$$\mathcal{W} = Q C Q^*, \quad (7)$$

where $C \in \mathbb{C}^{n \times n}$ is the Cauchy matrix with entries

$$C_{ij} = -\frac{1}{\lambda_i + \lambda_j^*}.$$

(ii) For all $k = 1, \dots, n$,

$$\frac{1}{\|Q^{-1}\|^2} \lambda_k(C) \leq \lambda_k(\mathcal{W}) \leq \|Q\|^2 \lambda_k(C). \quad (8)$$

Proof: (i) The decomposition (7) is a classic result and follows from the diagonalization of A via V and the explicit integral expression of the Gramian; see, e.g., [1, Lemma 9.4].

(ii) For any nonzero $x \in \mathbb{C}^n$, set $y := Q^* x$. Then

$$x^* \mathcal{W} x = x^* Q C Q^* x = y^* C y.$$

Moreover,

$$\frac{1}{\|Q^{-1}\|^2} x^* x \leq x^* Q Q^* x = y^* y \leq \|Q\|^2 x^* x,$$

which implies

$$\frac{1}{\|Q^{-1}\|^2} \frac{y^* \mathcal{C} y}{y^* y} \leq \frac{x^* \mathcal{W} x}{x^* x} \leq \|Q\|^2 \frac{y^* \mathcal{C} y}{y^* y}.$$

By the Courant-Fischer min-max characterization of eigenvalues [12, Theorem 4.2.6], this yields (8). ■

Building on Lemma 1, we next derive explicit spectral bounds for the controllability Gramian in terms of the eigenvalues of A .

Theorem 1: (Bounds on Gramian eigenvalues) Consider the system (1), and let $\{\lambda_i\}_{i=1}^n$ denote the eigenvalues of A . For any nonempty subset $S \subseteq \{1, \dots, n\}$ and $i \in S$, define

$$\theta_i(S) := \frac{1}{2|\operatorname{Re}(\lambda_i)|} \prod_{j \in S \setminus \{i\}} \left| \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j^*} \right|^2. \quad (9)$$

Then, for each $k = 1, \dots, n$, it holds

$$\lambda_k(\mathcal{W}) \geq c_{\min} \max_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \min_{i \in S} \theta_i(S), \quad (10)$$

$$\lambda_k(\mathcal{W}) \leq c_{\max} \min_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \max_{i \in T} \theta_i(\bar{T} \cup \{i\}), \quad (11)$$

where $c_{\min} := 1/(\|Q^{-1}\|^2 k)$ and $c_{\max} := \|Q\|^2(n-k+1)$ with Q defined in Lemma 1, and $\bar{T}$ is the complement of T in $\{1, \dots, n\}$.

Proof: We start with the lower bound. By Lemma 1(ii),

$$\lambda_k(\mathcal{W}) \geq \frac{1}{\|Q^{-1}\|^2} \lambda_k(\mathcal{C}).$$

Applying Cauchy's interlacing theorem [12, Theorem 4.3.17] gives

$$\lambda_k(\mathcal{C}) \geq \max_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \lambda_k(\mathcal{C}[S]),$$

so that

$$\lambda_k(\mathcal{W}) \geq \frac{1}{\|Q^{-1}\|^2} \max_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \lambda_k(\mathcal{C}[S]).$$

For any positive definite matrix M , $\lambda_{\min}(M)^{-1} \leq \operatorname{tr}(M^{-1})$, hence

$$\lambda_k(\mathcal{W}) \geq \frac{1}{\|Q^{-1}\|^2} \max_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \frac{1}{\operatorname{tr}(\mathcal{C}[S]^{-1})}.$$

Using $\operatorname{tr}(M^{-1}) \leq k \max_i (M^{-1})_{ii}$ for a $k \times k$ invertible matrix M ,

$$\lambda_k(\mathcal{W}) \geq \frac{1}{\|Q^{-1}\|^2 k} \max_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \frac{1}{\max_{i \in S} (\mathcal{C}[S]^{-1})_{ii}}.$$

Finally, by using Lemma 2(ii) in Appendix A with $x_i = -\lambda_i$, $y_j = -\lambda_j^*$, we obtain an explicit expression for $(\mathcal{C}[S]^{-1})_{ii}$ which yields (10).

For the upper bound, consider $\mathcal{C}^{-1}$. Applying again Cauchy's interlacing theorem,

$$\lambda_{n-k+1}(\mathcal{C}^{-1}) \geq \max_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \lambda_{n-k+1}((\mathcal{C}^{-1})[T]).$$

Using again the fact that for any positive definite matrix M , $\lambda_{\min}(M)^{-1} \leq \operatorname{tr}(M^{-1})$,

$$\begin{aligned} \lambda_{n-k+1}(\mathcal{C}^{-1}) &\geq \max_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \frac{1}{\operatorname{tr}(((\mathcal{C}^{-1})[T])^{-1})} \\ &= \frac{1}{\min_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \operatorname{tr}(((\mathcal{C}^{-1})[T])^{-1})}. \end{aligned}$$

Since $\lambda_k(\mathcal{C}) = 1/\lambda_{n-k+1}(\mathcal{C}^{-1})$, we obtain

$$\lambda_k(\mathcal{W}) \leq \|Q\|^2 \lambda_k(\mathcal{C}) \leq \|Q\|^2 \min_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \operatorname{tr}(((\mathcal{C}^{-1})[T])^{-1}),$$

where the first inequality follows from Lemma 1(ii). Applying the trace bound again,

$$\lambda_k(\mathcal{W}) \leq \|Q\|^2(n-k+1) \min_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \max_{i \in T} ((\mathcal{C}^{-1})[T])^{-1}_{ii}.$$

Finally, substituting in the above inequality the explicit formula of Lemma 2(iii) in Appendix A with $x_i = -\lambda_i$, $y_j = -\lambda_j^*$ yields (11). ■

It is worth noting that the lower and upper bounds in Theorem 1 have the same basic product structure. Indeed, both involve terms of the form $\theta_i(\cdot)$ evaluated on sets of cardinality k , and therefore both contain products of exactly $k-1$ eigenvalue-dependent factors

$$\left| \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j^*} \right|^2 < 1.$$

Thus, when these factors have a regular behavior, the bounds naturally lead to exponential-type decay estimates. The remaining prefactors $c_{\min}$ and $c_{\max}$ depend on Q , and hence on the conditioning of the eigenvector matrix together with the modal input weights. When these conditioning effects are moderate, the product terms θ_i capture the dominant scaling behavior of the bounds.

As a direct consequence of Theorem 1, we next derive explicit bounds for the largest and smallest eigenvalues of the controllability Gramian.

Corollary 1: (Bounds on extremal Gramian eigenvalues) Consider the system (1), and let $\{\lambda_i\}_{i=1}^n$ denote the eigenvalues of A . Then:

(i) the maximum eigenvalue of $\mathcal{W}$ satisfies

$$\frac{1/\|Q^{-1}\|^2}{2 \min_i |\operatorname{Re}(\lambda_i)|} \leq \lambda_1(\mathcal{W}) \leq \frac{n\|Q\|^2}{2 \min_i |\operatorname{Re}(\lambda_i)|}, \quad (12)$$

where Q is defined in Lemma 1.

(ii) the minimum eigenvalue of $\mathcal{W}$ satisfies

$$\frac{1}{\|Q^{-1}\|^2 n} \theta_{\min} \leq \lambda_n(\mathcal{W}) \leq \|Q\|^2 \theta_{\min}, \quad (13)$$

where $\theta_{\min} := \min_{i=1,\dots,n} \theta_i(\{1, \dots, n\})$ and $\theta_i(\cdot)$ is defined in (9).

Proof: The result follows directly from Theorem 1 by substituting $k = 1$ for the upper and lower bounds on $\lambda_1(\mathcal{W})$, and $k = n$ for those on $\lambda_n(\mathcal{W})$. ■

Remark 1: (Computational considerations) The bounds in Theorem 1 require an optimization over all index subsets of a given cardinality, which generally entails combinatorial complexity. However, since each inequality in the theorem holds for every admissible subset S or T , the bounds remain valid even when these subsets are fixed rather than optimized. In this case, their evaluation becomes straightforward, as it involves computing at most n terms that depend only on the eigenvalues of A . This reformulation simplifies the numerical evaluation of the bounds and makes their dependence on the eigenvalues of A through the factors θ_i more transparent. ▲

Beyond offering computationally tractable estimates for the Gramian eigenvalues, the bounds in Theorem 1 also provide analytical insight into how these eigenvalues decay as the system dimension increases, as we illustrate next.

Example 1: (Gramian eigenvalues of a tridiagonal chain) We consider the system described by

$$A = \text{tridiag}(1, -2, 1) \in \mathbb{R}^{n \times n}, \quad b = e_1.$$

Fig. 1 compares the bounds of Theorem 1 with the numerically computed eigenvalues of the controllability Gramian, for $n = 10$ and $n = 50$. The blue curve represents the direct numerical computation of $\lambda_k(\mathcal{W})$. For $n = 50$, this computation becomes unreliable for moderate-to-large values of k , as reflected by the missing markers in the bottom plot. In contrast, the bounds remain computable.

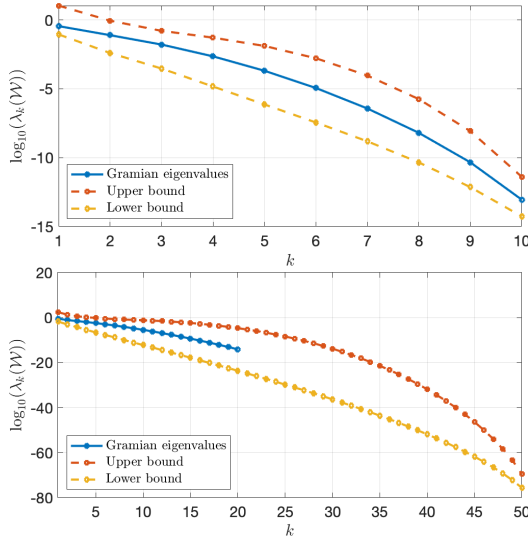


Fig. 1. Bounds on the eigenvalues of the controllability Gramian for the tridiagonal chain with $b = e_1$. The top and bottom plots correspond to $n = 10$ and $n = 50$, respectively. The bounds from Theorem 1 are evaluated using the fixed admissible choices $S = \{1, \dots, k\}$ and $T = \{k, \dots, n\}$. The blue markers show the eigenvalues obtained by computing the Gramian with MATLAB's `lyap`; missing markers indicate eigenvalues that are not reliably resolved numerically.

For this example, the eigenvalues of A are available in

closed form. This allows us to use the bounds above to derive analytical estimates of the scaling behavior of the Gramian eigenvalues. In particular, the eigenpairs of A are given by (see, e.g., [15, Ex. 7.2.5])

$$\lambda_i = -4 \sin^2\left(\frac{i\pi}{2(n+1)}\right), \quad V_{ij} = \sqrt{\frac{2}{n+1}} \sin\left(\frac{ij\pi}{n+1}\right). \quad (14)$$

Note that V is orthogonal, so that $\|Q\|^2 = \max_i d_i^2$, $1/\|Q^{-1}\|^2 = \min_i d_i^2$, with $d_i := (V^\top b)_i$. Moreover, from (14),

$$(V^\top b)_i^2 = \frac{2}{n+1} \sin^2\left(\frac{i\pi}{n+1}\right).$$

From $\sin x \geq \frac{2}{\pi}x$ for $x \in [0, \pi/2]$ and $\sin x \leq 1$ and the definition of $c_{\min}, c_{\max}$, it holds²

$$c_{\min} \geq \frac{8}{k(n+1)^3} \geq \frac{8}{n(n+1)^3}, \quad (15)$$

$$c_{\max} \leq \frac{2(n-k+1)}{n+1} \leq \frac{2n}{n+1}. \quad (16)$$

Upper bound. Assume $k \geq 2$. From (11), take $T = \{k, \dots, n\}$. The behavior of $\theta_i(\bar{T} \cup \{i\})$ for this choice is analyzed in Appendix B-1), where it is shown that, for all $i \in T$,

$$\theta_i(\bar{T} \cup \{i\}) \leq \frac{(n+1)^2}{8k^2} e^{-\rho(k-1)},$$

where $\rho := \frac{3}{8} \frac{k^2}{n^2}$. Therefore, from (11) and (16),

$$\begin{aligned} \lambda_k(\mathcal{W}) &\leq c_{\max} \max_{i \in T} \theta_i(\bar{T} \cup \{i\}) \\ &\leq C_{\max}(n) e^{-\rho k}, \end{aligned}$$

for some positive constant $C_{\max}(n)$ that depends polynomially on n .

Lower bound. Assume $k \geq 2$. Using (10) with $S = \{1, \dots, k\}$, in Appendix B-2) the following lower bound on each $\theta_i(S)$, $i \in S$, is derived

$$\theta_i(S) \geq \frac{1}{8} e^{-\eta(k-1)},$$

where $\eta := 4 \log(4\pi e)$. Hence, from (10) and (15),

$$\lambda_k(\mathcal{W}) \geq C_{\min}(n) e^{-\eta k},$$

for some positive constant $C_{\min}(n)$ that depends polynomially on n .

Asymptotic scaling. When k grows linearly with n , i.e. $k = cn$, $c \in (0, 1]$, it holds $\rho = \frac{3}{8} c^2$ is a positive constant independent of n .³ In this regime, the bounds above imply that

$$C_{\min}(n) e^{-\eta cn} \leq \lambda_k(\mathcal{W}) \leq C_{\max}(n) e^{-\rho cn},$$

that is, $\lambda_k(\mathcal{W})$ decays exponentially with n .

Observe also that when k grows sublinearly with n , the lower bound implies that $\lambda_k(\mathcal{W})$ cannot decay exponentially

²For $i = 1$, $\sin(\frac{\pi}{n+1}) \geq \frac{2}{n+1}$, giving $\min_i d_i^2 \geq \frac{8}{(n+1)^3}$, while $\max_i d_i^2 \leq \frac{2}{n+1}$ readily follows from $\sin x \leq 1$.

³One also checks that $\rho \leq \eta$ for all $c \in (0, 1]$. Indeed, $\rho \leq 3/8 \leq 4 \log(4\pi e) = \eta$.

in n , and thus the decay is at most subexponential as $n \rightarrow \infty$. In particular, when $k = \log n$, the lower bound yields

$$\lambda_k(\mathcal{W}) \geq C_{\min}(n)e^{-\eta \log n} = C_{\min}(n)n^{-\eta},$$

so that the decay is at most polynomial in n . $\diamond$

IV. SANDWICH BOUNDS ON HANKEL SINGULAR VALUES

As recalled in Section II, the Hankel singular values of system (1) are the square roots of the eigenvalues of the product of the controllability and observability Gramians. Because they depend directly on the spectrum of this product, one expects a close correspondence between the bounds derived above and those governing the Hankel singular values. This correspondence is formalized in the following result.

Theorem 2: (Bounds on Hankel singular values) Consider the system (1) and let $\{\sigma_k\}_{k=1}^n$ denote its Hankel singular values ordered so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$. Assume, in addition, that A has real eigenvalues. Let $V \in \mathbb{R}^{n \times n}$ be an eigenvector matrix of A , and define $d := V^{-1}b$, $f := cV$, and $r_i := d_i f_i$ for $i = 1, \dots, n$. If all r_i share the same sign, then, for each $k = 1, \dots, n$, it holds

$$\sigma_k \geq q_{\min} \max_{\substack{S \subseteq \{1, \dots, n\} \\ |S|=k}} \min_{i \in S} \theta_i(S), \quad (17)$$

$$\sigma_k \leq q_{\max} \min_{\substack{T \subseteq \{1, \dots, n\} \\ |T|=n-k+1}} \max_{i \in T} \theta_i(\overline{T} \cup \{i\}), \quad (18)$$

where $q_{\min} := r_{\min}/k$, $q_{\max} := r_{\max}(n - k + 1)$ with $r_{\min} := \min_i |r_i|$, $r_{\max} := \max_i |r_i|$, and $\theta_i(\cdot)$ is defined in Theorem 1.

Proof: By Lemma 1(i), and since A has real eigenvalues, the controllability Gramian admits the factorization

$$\mathcal{W} = Q\mathcal{C}Q^\top,$$

where $Q = V \text{diag}(V^{-1}b)$ and $\mathcal{C}$ is the Cauchy matrix with entries $\mathcal{C}_{ij} = -\frac{1}{\lambda_i + \lambda_j}$.

By adapting Lemma 1(i), we obtain a similar factorization for the observability Gramian,

$$\mathcal{O} = L^\top \mathcal{C}L,$$

where $L = \text{diag}(cV)V^{-1}$. Hence,

$$\mathcal{W}\mathcal{O} = V \text{diag}(V^{-1}b) \mathcal{C} R \mathcal{C} \text{diag}(cV)V^{-1},$$

where $R = Q^\top L^\top = \text{diag}(r_1, \dots, r_n)$.

Since $\text{diag}(V^{-1}b)$, $\text{diag}(cV)$ are invertible⁴, and $\mathcal{C}$ is positive definite, it follows that $\mathcal{W}\mathcal{O}$ is similar to

$$\mathcal{C}^{1/2} R \mathcal{C} R \mathcal{C}^{1/2},$$

where $\mathcal{C}^{1/2}$ denotes the principal (positive definite) square root of $\mathcal{C}$. Next, since, by assumption, all r_i share the same sign, there exists $\varepsilon \in \{\pm 1\}$ such that $R = \varepsilon |R|$, where $|R| = \text{diag}(|r_1|, \dots, |r_n|)$. Therefore,

$$\mathcal{C}^{1/2} R \mathcal{C} R \mathcal{C}^{1/2} = \mathcal{C}^{1/2} |R| \mathcal{C} |R| \mathcal{C}^{1/2} = (\mathcal{C}^{1/2} |R| \mathcal{C}^{1/2})^2.$$

⁴This follows from the standing controllability/observability assumptions.

Consequently, for all $k = 1, \dots, n$,

$$\sigma_k = \lambda_k(\mathcal{C}^{1/2} |R| \mathcal{C}^{1/2}).$$

Because $r_{\min} I \preceq |R| \preceq r_{\max} I$, it follows that

$$r_{\min} \lambda_k(\mathcal{C}) \leq \sigma_k \leq r_{\max} \lambda_k(\mathcal{C}).$$

The desired inequalities (17)-(18) then follow directly from the bounds on $\lambda_k(\mathcal{C})$ derived in the proof of Theorem 1. $\blacksquare$

The previous result shows that bounds similar to those obtained for Gramian eigenvalues also hold for the Hankel singular values, provided that A has real eigenvalues and the scalar coefficients r_i satisfy a same-sign condition. The meaning of this assumption is clarified below.

Remark 2: The quantities $r_i = d_i f_i$ in Theorem 2 are precisely the *residues* of the transfer function $G(s) = c(sI - A)^{-1}b$. Indeed, using the eigendecomposition $A = V\Lambda V^{-1}$, with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, we have

$$G(s) = cV(sI - \Lambda)^{-1}V^{-1}b = \sum_{i=1}^n \frac{r_i}{s - \lambda_i},$$

so that each r_i is the residue associated with the simple pole λ_i of $G(s)$. When all r_i are positive, the system is a *relaxation system*, that is, a strictly stable linear system whose transfer function is real, with real poles and positive residues. Such systems represent a broad class of dissipative or diffusive models, for example RC networks, thermal systems, and viscoelastic dynamics, whose impulse responses are nonnegative sums of decaying exponentials [10]. $\blacktriangle$

We next revisit the tridiagonal system of Example 1 and apply Theorem 2 to characterize the scaling of its Hankel singular values.

Example 2: (Hankel singular values of a tridiagonal chain) Let

$$A = \text{tridiag}(1, -2, 1) \in \mathbb{R}^{n \times n}, \quad b = e_1, \quad c = e_1^\top + \frac{1}{2}e_n^\top.$$

From the explicit eigenpair expressions of A in (14), it follows that

$$d_i = (V^\top e_1)_i = \sqrt{\frac{2}{n+1}} \sin\left(\frac{i\pi}{n+1}\right) \\ (e_n^\top V)_i = (-1)^{i+1} d_i,$$

so that

$$f_i = ((e_1^\top + \frac{1}{2}e_n^\top)V)_i = d_i \left(1 - \frac{1}{2}(-1)^i\right).$$

Therefore, with $\beta_i := 1 - \frac{1}{2}(-1)^i \in \{\frac{1}{2}, \frac{3}{2}\}$ and $m_i := \min\{i, n+1-i\}$,

$$r_i = d_i f_i = \frac{2}{n+1} \sin^2\left(\frac{\pi m_i}{n+1}\right) \beta_i > 0.$$

Thus the assumptions of Theorem 2 are satisfied.

Fig. 2 compares the bounds of Theorem 2 with the numerically computed Hankel singular values, for $n = 10$ and $n = 50$. The blue curve represents the direct numerical computation of σ_k . This computation becomes unreliable for moderate-to-large values of k , as indicated by the missing

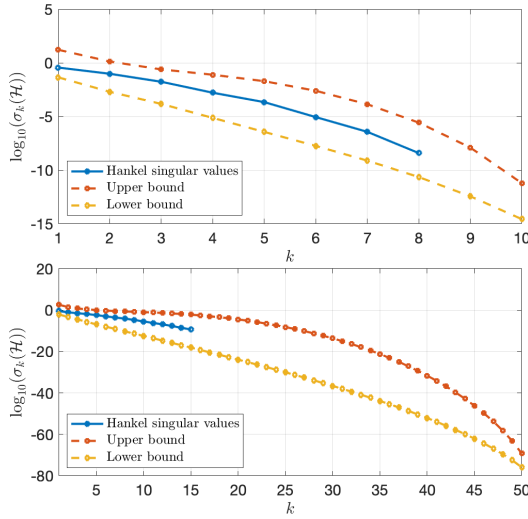


Fig. 2. Bounds on the Hankel singular values for the tridiagonal chain with $b = e_1$ and $c = e_1^\top + \frac{1}{2}e_n^\top$. The top and bottom plots correspond to $n = 10$ and $n = 50$, respectively. The bounds from Theorem 2 are evaluated using the fixed admissible choices $S = \{1, \dots, k\}$ and $T = \{k, \dots, n\}$. The blue markers show the Hankel singular values computed from the Gramians obtained using MATLAB's `lyap`; missing markers indicate singular values that are not reliably resolved numerically.

markers in the plots. In contrast, the bounds remain computable.

As in the previous example, the explicit spectral expressions allow us to derive closed-form decay estimates for the bounds. From $\sin x \geq \frac{2}{\pi}x$ and $\sin x \leq x$ on $[0, \pi/2]$ and the definition of $q_{\min}$, $q_{\max}$, we obtain

$$q_{\min} \geq \frac{4}{k(n+1)^3} \geq \frac{4}{n(n+1)^3},$$

$$q_{\max} \leq \frac{3(n-k+1)}{n+1} \leq \frac{3n}{n+1}.$$

By applying Theorem 2, we obtain results analogous to those in Example 1. In particular, when $k = cn$ with $c \in (0, 1]$, the Hankel singular values σ_k decay exponentially with n . Conversely, when k grows sublinearly with n , the lower bound ensures a subexponential decay, and in the specific case $k = \log n$, the decay is at most polynomial in n . Since the Hankel singular values determine the optimal Hankel-norm approximation error through (6), these results also describe how the achievable model reduction error scales with the system dimension. $\diamond$

V. CONCLUSIONS

This work has derived explicit upper and lower bounds for the eigenvalues of the controllability and observability Gramians, as well as for the Hankel singular values of SISO LTI systems with diagonalizable state matrix. The proposed Cauchy-matrix formulation provides closed-form expressions that make the dependence of these spectral quantities on the state matrix eigenvalues explicit. The resulting bounds offer both computationally efficient estimates and analytical insight into the scaling laws that govern how the Gramian and Hankel spectra, and thus the achievable accuracy of optimal model reduction techniques, vary with the system dimension.

Future work will focus on extending these results to multi-input multi-output systems, on relaxing the additional assumptions in Theorem 2, and on deriving exact asymptotic decay rates, as recently obtained in [5] for the smallest eigenvalue of the controllability Gramian.

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APPENDIX

A. Auxiliary results on Cauchy matrices

We recall that, for a matrix $A \in \mathbb{R}^{n \times n}$, we denote by $A[P, Q]$ the submatrix obtained by selecting the rows of A indexed by P and the columns indexed by Q . When $P = Q$, we simply write $A[P]$; in this case, $A[P]$ is the principal submatrix of A indexed by P .

Lemma 2: (Principal submatrices of a Cauchy matrix) Let $C \in \mathbb{C}^{n \times n}$ be a Cauchy matrix with entries

$$C_{ij} = \frac{1}{x_i + y_j}, \quad x_i + y_j \neq 0, \quad x_i \neq x_j, \quad y_i \neq y_j \quad \text{for } i \neq j.$$

Let $P \subseteq \{1, \dots, n\}$, $|P| = k$, and $\bar{P} := \{1, \dots, n\} \setminus P$. Then:

- (i) $C[P]$ is a Cauchy matrix with entries

$$(C[P])_{ij} = \frac{1}{x_i + y_j}, \quad i, j \in P.$$

- (ii) $C[P]$ is invertible and, for $i, j \in P$,

$$(C[P]^{-1})_{ij} = (x_j + y_i) \prod_{\substack{\ell \in P \\ \ell \neq i}} \frac{x_j + y_\ell}{y_i - y_\ell} \prod_{\substack{\ell \in \bar{P} \\ \ell \neq j}} \frac{x_\ell + y_i}{x_j - x_\ell}.$$

- (iii) $(C^{-1})[P]$ is invertible and

$$((C^{-1})[P])^{-1} = D_x C[P] D_y,$$

where $D_x = \text{diag}(\alpha_i)_{i \in P}$, $\alpha_i = \prod_{\ell \in \bar{P}} \frac{x_i - x_\ell}{x_i + y_\ell}$, and $D_y = \text{diag}(\beta_i)_{i \in P}$, $\beta_i = \prod_{\ell \in \bar{P}} \frac{y_i - y_\ell}{x_\ell + y_i}$.

Proof: (i) Restricting C to the index set P gives

$$(C[P])_{ij} = \frac{1}{x_i + y_j}, \quad i, j \in P,$$

so $C[P]$ is again a Cauchy matrix with parameters $\{x_i\}_{i \in P}$ and $\{y_i\}_{i \in P}$.

(ii) Cauchy matrices with distinct parameters are nonsingular and admit an explicit inverse formula; see, e.g., [9, Fact 3.20.14]. Applying this formula to $C[P]$ gives the stated expression.

(iii) Since C and its principal submatrix $C[\bar{P}]$ are nonsingular Cauchy matrices, $(C^{-1})[P]$ is invertible. Indeed, by the Schur determinant formula,

$$\det((C^{-1})[P]) = \frac{\det(C[\bar{P}])}{\det(C)} \neq 0.$$

We now compute $((C^{-1})[P])^{-1}$. For $i, j \in P$, the inverse formula applied to C gives

$$(C^{-1})_{ij} = (x_j + y_i) \prod_{\substack{\ell=1 \\ \ell \neq i}}^n \frac{x_j + y_\ell}{y_i - y_\ell} \prod_{\substack{\ell=1 \\ \ell \neq j}}^n \frac{x_\ell + y_i}{x_j - x_\ell}.$$

Splitting the products over P and $\bar{P}$, we obtain

$$(C^{-1})_{ij} = (x_j + y_i) \prod_{\substack{\ell \in P \\ \ell \neq i}} \frac{x_j + y_\ell}{y_i - y_\ell} \prod_{\substack{\ell \in P \\ \ell \neq j}} \frac{x_\ell + y_i}{x_j - x_\ell} \\ \cdot \prod_{\ell \in \bar{P}} \frac{x_j + y_\ell}{y_i - y_\ell} \prod_{\ell \in \bar{P}} \frac{x_\ell + y_i}{x_j - x_\ell}.$$

By the definitions of α_j and β_i ,

$$\prod_{\ell \in \bar{P}} \frac{x_j + y_\ell}{y_i - y_\ell} \prod_{\ell \in \bar{P}} \frac{x_\ell + y_i}{x_j - x_\ell} = \frac{1}{\beta_i \alpha_j}.$$

Hence, by part (ii),

$$(C^{-1})_{ij} = \frac{1}{\beta_i \alpha_j} (C[P]^{-1})_{ij}.$$

Equivalently,

$$(C^{-1})[P] = D_y^{-1} C[P]^{-1} D_x^{-1}.$$

Since D_x and D_y are invertible, we conclude that

$$((C^{-1})[P])^{-1} = D_x C[P] D_y.$$

B. Bounds on $\theta_i(S)$ for the tridiagonal chain

This section provides the bounds for $\theta_i(S)$ used in Example 1. Recall that

$$\theta_i(S) = \frac{1}{2|\lambda_i|} \prod_{j \in S \setminus \{i\}} \left(\frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j} \right)^2, \quad \lambda_i = -4 \sin^2 \left(\frac{i\pi}{2(n+1)} \right),$$

and that $|\lambda_i|$ increases monotonically with i .

1) *Upper bound on $\theta_i(\bar{T} \cup \{i\})$:* Let $T = \{k, \dots, n\}$ and fix $i \in T$. Define $r_{ij} := \frac{|\lambda_j|}{|\lambda_i|} \in (0, 1)$ for $j < k$. Using $\log \frac{1-r}{1+r} = -2 \operatorname{atanh}(r)$, we have

$$\theta_i(\bar{T} \cup \{i\}) = \frac{1}{2|\lambda_i|} \prod_{j < k} \left(\frac{1 - r_{ij}}{1 + r_{ij}} \right)^2 \\ = \frac{1}{2|\lambda_i|} e^{-4 \sum_{j < k} \operatorname{atanh}(r_{ij})}.$$

Since $\operatorname{atanh}(x) = x + \frac{x^3}{3} + \frac{x^5}{5} \dots \geq x$ for $x \in [0, 1]$,

$$\theta_i(\bar{T} \cup \{i\}) \leq \frac{1}{2|\lambda_i|} e^{-4 \sum_{j < k} r_{ij}}. \quad (19)$$

We now lower bound the sum in the exponent. Let $\alpha := \frac{\pi}{2(n+1)}$. Then $|\lambda_\ell| = 4 \sin^2(\ell\alpha)$, and note that $\sin x \geq \frac{2}{\pi}x$ for $x \in [0, \frac{\pi}{2}]$ and $\sin x \leq 1$, which imply

$$|\lambda_\ell| \geq 4 \left(\frac{2}{\pi} \ell \alpha \right)^2 = \frac{4\ell^2}{(n+1)^2}, \quad |\lambda_\ell| \leq 4.$$

Hence, for every $j < k \leq i$,

$$r_{ij} = \frac{|\lambda_j|}{|\lambda_i|} \geq \frac{|\lambda_j|}{4} \geq \frac{j^2}{(n+1)^2}.$$

Summing over $j = 1, \dots, k-1$ and using $\sum_{j=1}^{k-1} j^2 = \frac{(k-1)k(2k-1)}{6}$,

$$\sum_{j < k} r_{ij} \geq \frac{1}{(n+1)^2} \frac{(k-1)k(2k-1)}{6}. \quad (20)$$

Insert (20) in (19) to obtain

$$\theta_i(\bar{T} \cup \{i\}) \leq \frac{1}{2|\lambda_i|} e^{-\frac{2}{3} \frac{(k-1)k(2k-1)}{(n+1)^2}}. \quad (21)$$

Finally, since $i \geq k$ and $|\lambda_i|$ increases with i ,

$$\frac{1}{2|\lambda_i|} \leq \frac{1}{2|\lambda_k|}, \quad |\lambda_k| = 4 \sin^2 \left(\frac{k\pi}{2(n+1)} \right) \geq \frac{4k^2}{(n+1)^2},$$

which implies

$$\frac{1}{2|\lambda_k|} \leq \frac{(n+1)^2}{8k^2}. \quad (22)$$

From (21) and (22), it follows that

$$\theta_i(\bar{T} \cup \{i\}) \leq \frac{(n+1)^2}{8k^2} e^{-\frac{2}{3} \frac{(k-1)k(2k-1)}{(n+1)^2}} \\ \leq \frac{(n+1)^2}{8k^2} e^{-\frac{3}{8} \frac{k^2}{n^2} (k-1)}$$

where in the last step we used that, for $k \geq 2$ and $n \geq 2$,

$$\frac{2}{3} \frac{k(2k-1)}{(n+1)^2} \geq \frac{3}{8} \frac{k^2}{n^2}.$$

2) *Lower bound on $\theta_i(S)$:* Let $S = \{1, \dots, k\}$ and $\alpha := \frac{\pi}{2(n+1)}$. Using the bound⁵

$$|\sin^2(\frac{\pi}{2}x) - \sin^2(\frac{\pi}{2}y)| \geq (x-y)^2, \quad x, y \in [0, 1],$$

we have for all i, j ,

$$|\sin^2(i\alpha) - \sin^2(j\alpha)| \geq \frac{1}{(n+1)^2} (i-j)^2.$$

From $\sin(x) \leq x$ for $x \geq 0$, it follows that

$$\left| \frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j} \right| = \frac{|\sin^2(i\alpha) - \sin^2(j\alpha)|}{\sin^2(i\alpha) + \sin^2(j\alpha)} \geq \frac{4}{\pi^2} \left(\frac{i-j}{i+j} \right)^2.$$

Therefore,

$$\prod_{j=1, j \neq i}^k \left(\frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j} \right)^2 \geq \left(\frac{4}{\pi^2} \right)^{2(k-1)} \prod_{j=1, j \neq i}^k \left(\frac{i-j}{i+j} \right)^4.$$

Now $\prod_{j \neq i} |i-j| = (i-1)!(k-i)!$, while for each $j \neq i$ we have $i+j \leq i+k \leq 2k$, hence

$$\prod_{j \neq i} (i+j) \leq (2k)^{k-1}.$$

⁵For $x, y \in [0, 1]$, let $a = \frac{\pi}{2}x$, $b = \frac{\pi}{2}y$, and $d = |a-b| = \frac{\pi}{2}|x-y|$. Since $a, b \in [0, \frac{\pi}{2}]$, we have $a+b \in [d, \pi-d]$, hence $|\sin^2 a - \sin^2 b| = |\sin(a+b) \sin(a-b)| \geq \sin^2 d = \sin^2(\frac{\pi}{2}|x-y|) \geq (x-y)^2$, using $\sin x \geq \frac{2}{\pi}x$ for $x \in [0, \frac{\pi}{2}]$.

Therefore,

$$\prod_{j=1, j \neq i}^k \left(\frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j} \right)^2 \geq \left(\frac{4}{\pi^2} \right)^{2(k-1)} \left(\frac{(i-1)!(k-i)!}{(2k)^{k-1}} \right)^4.$$

Using the inequality $(i-1)!(k-i)! \geq \frac{(k-1)!}{2^{k-1}}$,⁶ we obtain

$$\prod_{j=1, j \neq i}^k \left(\frac{\lambda_i - \lambda_j}{\lambda_i + \lambda_j} \right)^2 \geq \left(\frac{4}{\pi^2} \right)^{2(k-1)} \left(\frac{(k-1)!}{(4k)^{k-1}} \right)^4.$$

Finally, since $|\lambda_i| \leq 4$,

$$\theta_i(S) \geq \frac{1}{8} \left(\frac{4}{\pi^2} \right)^{2(k-1)} \left(\frac{(k-1)!}{(4k)^{k-1}} \right)^4.$$

Using Stirling's inequality $(k-1)! \geq \left(\frac{k-1}{e} \right)^{k-1}$ yields

$$\theta_i(S) \geq \frac{1}{8} \left(\frac{4}{\pi^2} \right)^{2(k-1)} \left(\frac{k-1}{4ek} \right)^{4(k-1)} \geq \frac{1}{8} \left(\frac{1}{4\pi e} \right)^{4(k-1)},$$

where in the last step we used $k-1 \geq k/2$ for $k \geq 2$.

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⁶The inequality follows from $\binom{k-1}{i-1} = \frac{(k-1)!}{(i-1)!(k-i)!} \leq 2^{k-1}$.